

Structural Modeling With Network Data

Empirical Methods — Week 20

Teaching deck draft

Week 20 question

- When is a model of network formation or network-mediated behavior needed for counterfactuals?
- Lecture 18: define and diagnose the graph.
- Lecture 19: identify effects when links create exposure, interference, or dependence.
- Lecture 20: model policies that change links, information flows, search, matching, or equilibrium behavior.

The central issue is not graph theory. It is whether the counterfactual changes the network.

When reduced-form network methods are not enough

Reduced-form exposure design:

$$Y_i = \alpha + \tau D_i(G, Z) + X_i' \beta + u_i$$

Structural policy object:

$$V(p) = \mathbb{E} [W_i (G(p), a_i(G(p), X, \varepsilon), X_i)]$$

- Reduced-form evidence can identify local exposure effects on the observed graph.
- It may not answer what happens when policy changes the graph itself.
- Examples: referral subsidies, platform ranking rules, team assignment, diffusion seeds, search access.
- The structural model earns its keep only if the counterfactual changes relationships or equilibrium behavior.

Network methods block map

Lecture	Main object	Failure if skipped
18. Curation	nodes, edges, weights, timing, boundaries	unclear estimand and fragile measurement
19. Causal inference	exposure, interference, dependence	peer correlations mistaken for effects
20. Structure	formation, behavior, counterfactuals	link-changing policies overclaimed

The three layers are complements. The structural model should inherit the graph discipline from Lecture 18 and the design discipline from Lecture 19.

Network formation vs behavior

Behavior on a network

$$a_i = \alpha + \beta x_i + \gamma \sum_j g_{ij} a_j + \delta \sum_j g_{ij} x_j + \varepsilon_i$$

- Conditions on observed links.
- Needs timing, equilibrium, and exposure logic.
- Useful for peer effects, diffusion, learning, and adoption.

Network formation

$$G_{ij} = \mathbf{1}\{U_{ij}(X_i, X_j, G_{-ij}, \eta_{ij}) \geq 0\}$$

- Explains why links exist.
- Allows policy to change links.
- Needs assumptions on opportunity, sorting, and strategic payoffs.

Structural peer effects and equilibrium

$$a_i = \alpha + \beta x_i + \gamma \sum_j g_{ij} a_j + \delta \sum_j g_{ij} x_j + \varepsilon_i$$

- The equation is familiar; the structural burden is the equilibrium interpretation.
- Are peer actions simultaneous or lagged?
- Is the graph fixed before actions, or chosen jointly with actions?
- Can multiple equilibria matter?
- Which variation separates peer effects from sorting and common shocks?

A coefficient is not yet a counterfactual model.

Search, referrals, and labor-market applications

$$\lambda_i(G) = \lambda_0 + \lambda_1 \sum_j G_{ij} q_j$$

- Links can carry vacancy information, screening signals, trust, reputation, or favoritism.
- Referral policies may change who forms links, who receives information, and how employers update beliefs.
- Matching policies can change feasible worker–firm pairs and congestion.
- Inequality can rise or fall depending on whether new links bridge opportunity gaps or reinforce segregation.

Labor anchors: Calvo-Armengol and Jackson; Dustmann et al.; Galenianos; Pallais and Sands.

Diffusion and equilibrium behavior

$$\Pr(a_{it} = 1) = \sigma \left(\alpha + X_i' \beta + \rho \sum_j g_{ij} a_{j,t-1} + \kappa s_{it} \right)$$

- Diffusion models study how information, technologies, practices, and beliefs spread.
- Relevant for training, management practices, job-search norms, platform information, and wage knowledge.
- Equilibrium response matters when adoption changes congestion, incentives, or search behavior.
- Validation requires adoption paths, not only final adoption rates.

Estimation, computation, and validation

$m(\theta) = \mathbb{E}_\theta[\text{degree, clustering, homophily, wage covariance across links, job transitions, referral yields}]$

- Use moments or likelihood objects that discipline the counterfactual primitives.
- Formation moments: degree, homophily, clustering, link persistence.
- Behavior moments: outcomes conditional on links, transitions, adoption paths, referral yields.
- Computation often uses simulation, local statistics, sparse approximations, or separable blocks.
- Validation should compare fixed-network and endogenous-link counterfactuals.

Policy counterfactual discipline

Policy changes	Need to model	Validation target
Existing-link information New referral links	behavior on fixed graph formation and referral yields	outcome response by exposure degree, homophily, callback rates
Platform ranking	matching and congestion	search sets, applications, matches
Training seeds	diffusion and equilibrium re- sponse	adoption timing and spillovers
Team assignment	peer groups and sorting	group composition and produc- tivity

The policy defines the necessary structure.

Network Methods block summary

- **Where the literature is now:** richer administrative and platform networks; sharper exposure and interference designs; more structural formation, search, referral, and diffusion models.
- **Key open questions:** endogenous links, dynamic networks, privacy-preserving network data, reduced-form discipline for structural counterfactuals, and distributional effects of network policies.
- **Research opportunities:** referral hiring, internal labor markets, coworker learning, manager networks, platform matching, school-to-work contacts, and local opportunity networks.

The frontier is not bigger graphs. It is better alignment between mechanism, data, design, model, and counterfactual.

Reproduce

- synthetic workers
- dyadic opportunities
- link formation model
- network moments
- behavior equation

Diagnose

- topology fit
- behavior fit
- moment relevance
- fixed vs endogenous
- validation memo

Transfer

- referral search
- cross-group links
- match access
- inequality
- policy memo

Takeaways

- Structural network models are useful when policy changes links, information flows, matching sets, or equilibrium behavior.
- Behavior-on-network models and formation models answer different questions.
- Endogenous links make assumptions and validation burdens heavier.
- Moments must be chosen for the counterfactual, not for convenience.
- Reduced-form network evidence remains valuable as discipline for structural claims.
- The Network Methods block gives a sequence: define, identify, then model.